TA Slow Session

10 October 2026 22:52

Statistics for ML

CLT Practice Questions

Questions with worked solutions (TA copy)

Part A — Questions

Standard normal values $\Phi(z) = P(Z \leq z)$

z	0.49	0.52	1.28	1.54	1.59	1.60	2.36	3.51
$\Phi(z)$	0.6879	0.6985	0.8997	0.9382	0.9441	0.9452	0.9909	0.9998

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Question 1. Understanding the Central Limit Theorem

Suppose $X_1, X_2, \ldots, X_n$ are independent and identically distributed (i.i.d.) random variables with mean μ and standard deviation σ. Let

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

represent the sample mean. According to the Central Limit Theorem (CLT), which of the following statements are correct?

(a) The distribution of each random variable X_i is approximately symmetric around μ.

(b) For large n, the distribution of the sample mean $\bar{X}$ is approximately symmetric around μ, even if the X_i are skewed.

(c) For large n, the sample mean $\bar{X}$ approximately follows a normal distribution.

(d) For large n, $(\bar{X} - \mu)/\sigma$ approximately follows a standard normal distribution.

$X_i \sim$ iid $\quad F(\mu, \sigma) \quad \bar{X} = \frac{1}{n}\sum X_i$

$X_i \sim$ skewed

$\bar{X} \sim N(\mu, \sigma^2/n) \quad \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \sim N(\mu', \sigma')$; X_i iid

$E(\bar{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n} E\left(\sum X_i\right)$ $X_i \sim (\mu, \sigma^2)$

$= \frac{1}{n} E(X_1 + X_2 + \ldots + X_n)$

$= \frac{1}{n}[E(X_1) + E(X_2) + \ldots + E(X_n)] = \frac{n\mu}{n} = \mu$

$Var(\bar{X}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)$ ~ indep

$Var(aX+b) = a^2 Var X + b$

$= \frac{1}{n^2}\sum_{i=1}^{n} Var(X_i) = \frac{n \times \sigma^2}{n^2} = \frac{\sigma^2}{n}$

$\bar{X} \sim N(\mu, \sigma^2/n)$ $\quad \sigma_{\bar{x}} = \sigma/\sqrt{n} \rightarrow$ Standard error of

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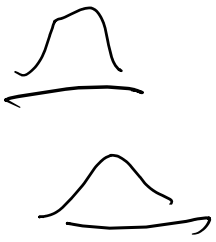
error of the mean

$Var x_1 = a \quad Var x_3 = b \quad Var x_{k-1} = d$

$a = b = d$ $\quad (x_1, x_2, \ldots, x_n) \sim$ indep (iid) identically distributed

$E(x_i) = \mu \; \forall \; i$

$Var(x_i) = \sigma^2 \; \forall \; i$



$h_1, h_2, h_3, \ldots, h_n \rightarrow$ not independent

$h_1, h_2, \ldots$

$X_i \sim F[\mu, \sigma^2]$

$\mu_1 = \mu_2 = \ldots = \mu_n = \mu$

$Var(h_1) = Var(h_2) = \ldots = Var(h_\alpha) = \sigma^2$

$\bar{X} \sim N(\mu, \sigma^2/n)$

$Var X = E[X - (E(X))]^2$

$\bar{X} \sim N(\mu, \sigma^2/n)$

$z = \dfrac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$\dfrac{\bar{x} - \mu}{\sigma} \sim$

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Solution 1. Understanding the CLT

Correct options: **(b) and (c)**.

- (a) False. The CLT does not require the individual X_i to be symmetric; they may be skewed.
- (b) True. For large n, $\bar{X}$ is approximately normal, hence approximately symmetric about μ, whatever the shape of the X_i.
- (c) True. The CLT says

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \approx N(0,1), \qquad \text{so} \qquad \bar{X} \approx N\left(\mu, \tfrac{\sigma^2}{n}\right).$$

- (d) False. The standardization must divide by the standard error $\sigma/\sqrt{n}$, not by σ.

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Question 2. Insurance Claims

An insurance company has 25,000 automobile policyholders. The yearly claim of a policyholder is a random variable with mean 320 and standard deviation 540. Assuming the yearly claims are independent and identically distributed, approximate the probability that the total yearly claim exceeds 8.3 million.

$n = 25,000$

$X_i \sim (320, 540^2)$

$P(X_1 + X_2 + \dots + X_n \geq 8.3 \times 10^6)$

$S = X_1 + X_2 + \dots + X_n$

$E(S) = E(X_1 + X_2 + \dots + X_n)$

$= E(X_1) + E(X_2) + \dots + E(X_n)$

$= n \times E(X_i)$

$= 25,000 \times 320$

$= 8 \times 10^6$

$Var(S) = Var(X_1 + X_2 + \dots + X_n)$

$= n\,Var(X_i)$

$= 25000 \times 540^2$

$\sigma_S = \sqrt{540^2 \times 25000}$

$= 540\sqrt{25000} \simeq 85,381$

$S \sim N(8 \times 10^6, (85,381))$

$P(S > 8.3 \times 10^6) = P\left(Z > \frac{8.3 \times 10^6 - 8 \times 10^6}{85,381}\right)$

α

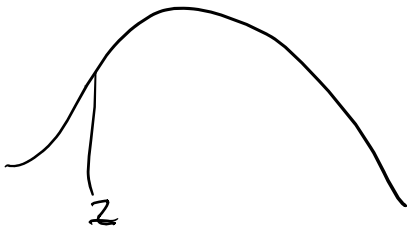
$P(Z > \alpha) = 1 - \Phi(\alpha)$

$\Phi(\alpha) = F_x(\alpha)$ $\quad \alpha = (3.51)$

$X \sim N(\mu, \sigma^2)$ $\quad Z \sim \frac{X - \mu}{\sigma} \sim N(0, 1)$

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Fl = —



= 0.00022

$$\alpha = \frac{S - \mu}{\sigma} = \frac{8.3\times10^6 - 8\times10^6}{85,381} = \alpha = 3.51$$

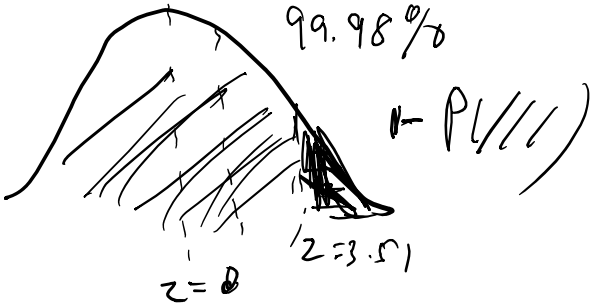
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$\Phi(z)$	0.6879	0.6985	0.8997	0.9382	0.9441	0.9452	0.9909	0.9998

$z = 3.51$

$P(S > 8.3\times10^6) = 1 - P(z \le 3.51)$

=



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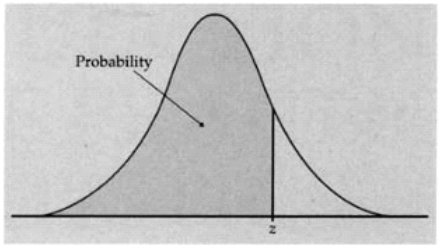


Table entry for z is the area under the standard normal curve to the left of z.

TABLE A Standard normal probabilities (*continued*)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

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SOLUTION Let X denote the total yearly claim. Number the policy holders, and let X_i denote the yearly claim of policy holder i. With $n = 25{,}000$, we have from the central limit theorem that $X = \sum_{i=1}^{n} X_i$ will have approximately a normal distribution with mean $320 \times 25{,}000 = 8 \times 10^6$ and standard deviation $540\sqrt{25{,}000} = 8.5381 \times 10^4$. Therefore,

$$P\{X > 8.3 \times 10^6\} = P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{8.3 \times 10^6 - 8 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$= P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{.3 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$\approx P\{Z > 3.51\} \qquad \text{where } Z \text{ is a standard normal}$$

$$\approx .00023$$

Thus, there are only 2.3 chances out of 10,000 that the total yearly claim will exceed 8.3 million. ■

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Question 3. Weight Supported by a Bridge

Civil engineers believe that W, the amount of weight (in units of 1,000 pounds) that a certain span of a bridge can withstand without structural damage, is normally distributed with mean 400 and standard deviation 40. The weight of a car (in the same units) is a random variable with mean 3 and standard deviation 0.3. Assume the car weights are independent of one another and of W. Let S_n be the total weight of n cars on the span.

(a) Find the mean and standard deviation of S_n.

(b) Structural damage occurs when $S_n > W$. Let $D = S_n - W$. Find the mean and standard deviation of D, and express $P(\text{damage})$ using Φ.

(c) How many cars must be on the span for the probability of structural damage to exceed 0.1? (Use $P(Z < 1.28) = 0.90$.)

(a) $W \sim N(400, 40^2)$; $X_i \sim F(3, 0.3^2)$

$S_n = X_1 + X_2 + \ldots + X_n \sim N(n\mu, n\sigma^2)$ $X_i \perp W$, $X_i \perp X_j$

$S_n \sim N(n\mu, n\sigma^2)$ $\sigma_{X_i} = 0.3$ where $i \neq j$

$n\mu = n \times 3$; $\sigma_{S_n} = \sqrt{n\sigma^2}$

$= \sigma\sqrt{n}$

$= 0.3\sqrt{n}$

(b) $S_n > W$

$P(\text{damage}) = P(S_n > W) = P(S_n - W > 0)$

If X_1, X_2 are normal RV then $X_1 + X_2$ is also normal

$D = S_n - W$ $E(D) = E(S_n) - E(W)$

$= 3n - 400$

$\text{Var}(D) = \text{Var}(S_n) + \text{Var}(W)$

$= 0.09n + 1600$

$D \sim N(3n - 400, 0.09n + 1600)$

$\sigma_D = \sqrt{0.09n + 1600}$

$P(\text{Damage}) = P(D > 0)$

$= P(\ldots > 400 - 3n)$

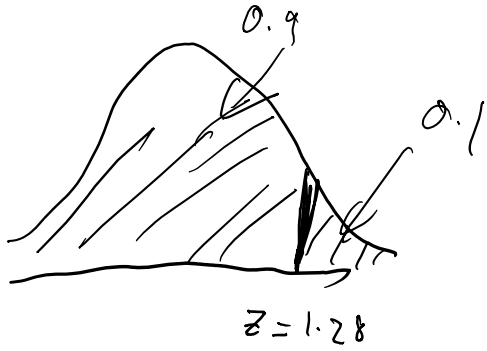
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$$= P\left(Z \geq \frac{400 - 3n}{\sqrt{6.09n + 1600}} \right)$$

$$=$$

(c) $P_n > 0.1$

$$P(Z < 1.28) = 0.9$$



$$\frac{400 - 3n}{\sqrt{1600 + 0.09n}} < 1.28$$

$$(400 - 3n)^2 = 1.28^2 (1600 + 0.09n)$$

$$n = 116, 117 \qquad n \geq 117$$

EXAMPLE 6.3b Civil engineers believe that W, the amount of weight (in units of 1,000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1,000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1?

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SOLUTION Let P_n denote the probability of structural damage when there are n cars on the bridge. That is,

$$\begin{aligned} P_n &= P\{X_1 + \cdots + X_n \geq W\} \\ &= P\{X_1 + \cdots + X_n - W \geq 0\} \end{aligned}$$

where X_i is the weight of the ith car, $i = 1, \ldots, n$. Now it follows from the central limit theorem that $\sum_{i=1}^{n} X_i$ is approximately normal with mean $3n$ and variance $.09n$. Hence, since W is independent of the $X_i, i = 1, \ldots, n$, and is also normal, it follows that $\sum_{i=1}^{n} X_i - W$ is approximately normal, with mean and variance given by

$$E\left[\sum_{1}^{n} X_i - W\right] = 3n - 400$$

$$\text{Var}\left(\sum_{1}^{n} X_i - W\right) = \text{Var}\left(\sum_{1}^{n} X_i\right) + \text{Var}(W) = .09n + 1{,}600$$

Therefore, if we let

$$Z = \frac{\sum_{i=1}^{n} X_i - W - (3n - 400)}{\sqrt{.09n + 1{,}600}}$$

then

$$P_n = P\left\{Z \geq \frac{-(3n - 400)}{\sqrt{.09n + 1{,}600}}\right\}$$

where Z is approximately a standard normal random variable. Now $P\{Z \geq 1.28\} \approx .1$, and so if the number of cars n is such that

$$\frac{400 - 3n}{\sqrt{.09n + 1{,}600}} \leq 1.28$$

or

$$n \geq 117$$

then there is at least 1 chance in 10 that structural damage will occur. ■

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Question 4. College Admissions

The ideal size of a first-year class at a particular college is 150 students. Based on past experience, on average only 30% of those accepted for admission will actually attend. The college therefore approves 450 applications. Assume each approved applicant decides independently whether to attend.

Compute the probability that more than 150 first-year students attend the college.

Use Indicator rv (Bernoulli rv)

Let X be no. of students

$X_i \xrightarrow{0.3} 1$

$\xrightarrow{0.7} 0$

$X = X_1 + X_2 + X_3 + \dots + X_n$

$E(X) = 450 \times 0.3 = 135$

$\text{Var}(X) = 450 \times 0.3 \times 0.7 = \frac{21}{100} \times 450$ (3/10, 7/10)

$\sigma_X = \sqrt{\frac{21 \times 45}{10}} = 9.72$

$P(X > 150) = P\left(Z > \frac{150 - 135}{9.72}\right)$

$P\left(Z > \frac{15}{9.72}\right)$

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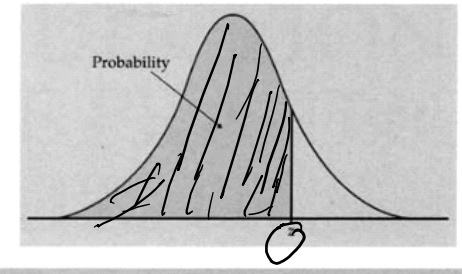


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0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
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0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	[illegible]	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

$= P(z > 1.54) = 1 - P(z \leq 1.54)$

$= 1 - 0.9382$

$= 0.0618$

EXAMPLE 6.3c The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

SOLUTION Let X denote the number of students that attend; then assuming that each accepted applicant will independently attend, it follows that X is a binomial random

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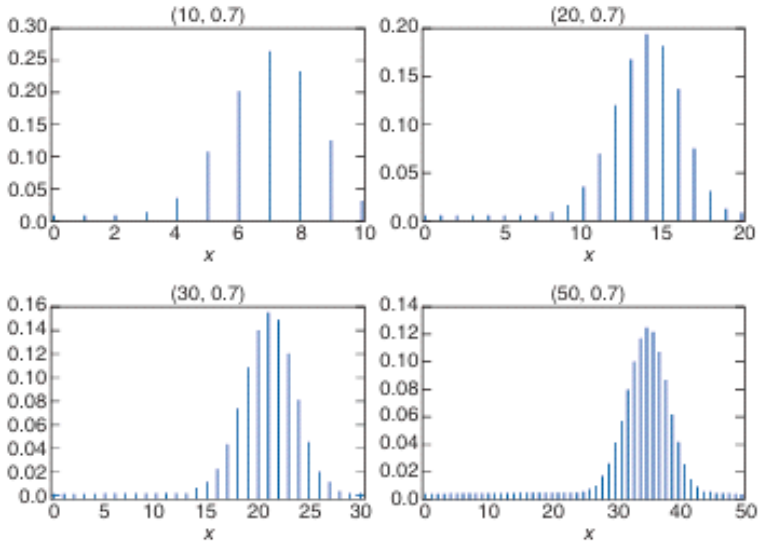


FIGURE 6.3 *Binomial probability mass functions converging to the normal density.*

variable with parameters $n = 450$ and $p = .3$. Since the binomial is a discrete and the normal a continuous distribution, it is best to compute $P\{X = i\}$ as $P\{i-.5 < X < i+.5\}$ when applying the normal approximation (this is called the continuity correction). This yields the approximation

$$P\{X > 150.5\} = P\left\{\frac{X - (450)(.3)}{\sqrt{450(.3)(.7)}} \geq \frac{150.5 - (450)(.3)}{\sqrt{450(.3)(.7)}}\right\}$$
$$\approx P\{Z > 1.59\} = .06$$

Hence, only 6 percent of the time do more than 150 of the first 450 accepted actually attend. ■

Slow Session Page 14 This file is meant for personal use by parikshithrnayak@gmail.com only. Sharing or publishing the contents in part or full is liable for legal action.

Question 5. Snowfall and Salt

A highway department has enough salt to handle a total of 80 inches of snowfall. Suppose the daily amount of snow has a mean of 1.5 inches and a standard deviation of 0.3 inches.

(a) Approximate the probability that the salt on hand will suffice for the next 50 days.

(b) What assumption did you make in solving part (a)?

(c) Do you think this assumption is justified? Explain briefly.

$X_i \sim F(1.5, 0.3^2)$

$S_{50} = X_1 + X_2 + \dots + X_{50}$

$N \sim (75, 0.3^2 \times 50)$

$\text{Var}(S_{50}) = 50\,\text{Var}\,X_i$

$\sigma_{S_{50}} = 0.3\sqrt{50}$

$E(X) = 50 \times 1.5$

$\sigma_X = \sqrt{0.3^2 \times 56}$

$= 0.3\sqrt{56}$

Answer : (a) 0.991

(b) iid assumption

(c) $\text{Var}(X+Y) = \text{Var}\,X + \text{Var}\,Y + 2\,\text{Cov}(X,Y)$

$\text{Var}_0(S) \geq \text{Var}(S)$

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Solution 5. Snowfall and Salt

Let X_i be the snowfall on day i and $S_{50} = \sum_{i=1}^{50} X_i$. The salt suffices if $S_{50} \leq 80$.

(a) Assuming the X_i are i.i.d.,

$$E[S_{50}] = 50(1.5) = 75, \qquad \sigma_{S_{50}} = \sqrt{50}\,(0.3) \approx 2.121,$$

$$P(S_{50} \leq 80) \approx \Phi\left(\frac{80 - 75}{2.121}\right) = \Phi(2.36) \approx \boxed{0.991}.$$

(b) We assumed the daily snowfall amounts are independent and identically distributed, with mean 1.5 and standard deviation 0.3.

(c) Not necessarily. Weather systems persist, so snowfall on consecutive days is likely positively correlated, and the mean and spread vary by season. Positive correlation makes the total more variable than $\sqrt{50}\,(0.3)$ suggests, so the true probability of running short is higher than this calculation indicates.

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Question 6. Repeated Die Rolls

A fair six-sided die is rolled repeatedly until the total of all rolls exceeds 500. Let N be the number of rolls required, and let S_{140} be the total of the first 140 rolls.

(a) Explain why $\{N > 140\}$ is the same event as $\{S_{140} \leq 500\}$.

(b) Approximate $P(N > 140)$.

$P(N>140) = P(S_{140} \leq 500)$

N = No. of rolls $\quad N>140 \rightarrow$ 140 rolls $\quad$ 120 roll exceed

$(N>140) \Rightarrow (S_{140} \leq 500)$

X_i be a rv $\rightarrow [1, 6]$ $\quad 1+2+\cdots+n = \frac{n(n+1)}{2}$

$E(X_i) = \frac{1}{6}\sum_{k=1}^{6} k = \frac{1}{6} \times \frac{6\times 7}{2} = 3.5$

$\text{Var}\, X_i = \frac{35}{12}$ $\quad n = 140$ $\quad$ CLT $N(n\mu, n\sigma^2)$

$n \to F(3.5, 35/12) \rightarrow$

$E(S) = E(X_1 + \cdots + X_n)$

$n\, F(3.5, 35/12) \Rightarrow N\left(3.5\times 140, \frac{35}{12}\cdot 140\right)$ $\quad 6\times 20$

$\sigma = \sqrt{\frac{35}{12}\times 140}$ $\quad N(\mu, \sigma^2)$

$\sim N(490, 420)$ $\quad \sigma = \sqrt{420} \approx 20.5$

$P(S_{140} \leq 500) = P\left(z \leq \frac{500-490}{\cdot}\right)$

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$$P(S_{100} \leq 500) = P\left(Z \leq \frac{500 - 490}{20.5}\right) \beta$$

$$P(Z \geq \alpha) = 1 - P(Z \leq \alpha)$$

$$= \Phi(\beta)$$

$$E(x_i) = 7/2 \qquad Var(x_i) = 35/12$$

$$Var(x_i) = E(x_i^2) - (E(x_i))^2 \rightarrow$$

$$E(x_i^2) = \frac{1}{6} \times 1^2 + \frac{1}{6} \times 2^2 + \ldots + \frac{1}{6} \times 6^2$$

$$= \frac{1}{6} \sum_{k=1}^{6} k^3 = \frac{1}{6} \times \frac{k(k+1)(2k+1)}{6}$$

$$= \frac{1}{6} \times \frac{6 \times 7 \times 13}{6} = \frac{91}{7}$$

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Solution 6. Repeated Die Rolls

(a) The running total only increases. So more than 140 rolls are needed exactly when the total after 140 rolls has not yet exceeded 500: $\{N > 140\} = \{S_{140} \le 500\}$.

(b) For one roll, $\mu = 3.5$ and $\sigma^2 = 35/12$. Hence

$$E[S_{140}] = 140(3.5) = 490, \qquad \sigma_{S_{140}} = \sqrt{140 \cdot \tfrac{35}{12}} \approx 20.21.$$

- **Plain CLT:** $P(S_{140} \le 500) \approx \Phi\left(\dfrac{500 - 490}{20.21}\right) = \Phi(0.49) \approx 0.688.$
- **With continuity correction:** $\Phi\left(\dfrac{500.5 - 490}{20.21}\right) = \Phi(0.52) \approx \boxed{0.70}.$

The exact value is 0.698.

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